# The Scale-Dual Subgrid Model of 3D Navier–Stokes: Its True Quadratic Invariant, Its Identity with the Modified Leray-$\alpha$ Model, and What Lean 4 Checks

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**Abstract**

We study the scale-dual subgrid model $\partial_t u = -P[(u\cdot\nabla)(M_{\alpha'}u)] + \nu\Delta u$ on $\mathbb{T}^3$, where the Fourier multiplier of $M_{\alpha'}$ is $m(k) = k_{\text{eff}}/|k| = \min(1, 1/(\alpha'|k|^2))$ (piecewise) or $m(k) = 1/(1+\alpha'|k|^2)$ (smooth). Edition 1 of this paper claimed that this model keeps the kinetic energy $E = \frac{1}{2}\sum|\hat{u}_k|^2$ non-increasing, and from that derived a global enstrophy ceiling and the absence of blow-up. *That claim is false*, and edition 2 corrects it openly. (i) Exact rational arithmetic on a single divergence-free Fourier triad gives $dE/dt = +6/5$ (piecewise) and $+32/15$ (smooth) at $\alpha' = 1/4$, and, at the paper's own $\alpha' = 1/25$, $\nu = 1/100$, a triad whose total $dE/dt$ is positive for every amplitude above $169/5$. (ii) The quantity the nonlinearity actually conserves is $Q = \frac{1}{2}\sum m(k)|\hat{u}_k|^2$. We prove in Lean 4, for every finite Galerkin truncation with a skew triad tensor, that $Q$ has zero nonlinear transfer, that $dQ/dt = -\nu\sum m\lambda|\hat{u}|^2 \le 0$ along every trajectory, and that $L^2$ conservation holds for all such tensors *if and only if* $m$ is constant. (iii) The smooth model is exactly the modified Leray-$\alpha$ (ML-$\alpha$) model of Ilyin, Lunasin and Titi [6], not the Leray-$\alpha$ model edition 1 named; $Q$ is its known energy. (iv) The final bound of edition 1, $\Omega_{\text{eff}} \le E_0/\alpha'$, survives, but only through $Q$: $\Omega_{\text{eff}} \le Q/\alpha' \le Q_0/\alpha' \le E_0/\alpha'$, Lean-checked for both symbols, the smooth case via $m_{\text{ML}} \le m_{\text{pw}}$ rather than by a single theorem. Because $\Omega_{\text{eff}}$ is not the enstrophy of $u$, it does not exclude blow-up. A pseudospectral Galerkin ensemble (299 runs, with positive and negative controls) reproduces the exact rationals to $10^{-15}$ and conserves $Q$ to $10^{-10}$ in the inviscid limit. It also shows the following about $E(t)$. At the paper's $(\alpha', \nu) = (1/25, 1/100)$ and the initially pre-registered amplitude $E_0 = 1/2$, $E(t) \le E_0$ held in all 10 seeds. At $E_0 = 2$, 8 and 32, the piecewise model exceeded $E_0$ in 5 of 5 seeds (mean $\max_t E/E_0 = 1.034$, 1.356, 1.767), while the Navier–Stokes control never did. At $E_0 = 32$ the excess persists under $dt \to dt/2$ (relative change $3.5\times10^{-7}$) and $N = 32^3 \to 64^3$, where it grows to $\approx 3.0$. The edition-1 inequality $E(t) \le E_0$ is therefore refuted by the pre-registered criterion, but the size of the excess is not converged in $N$. The Lean gate passes 13/13 declarations in `PaperB_EnergyBudget.lean` and 12/12 in `PaperB_ScaleDualEd2.lean`, with axioms exactly {`propext`, `Classical.choice`, `Quot.sound`}.

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# 1 Corrections to edition 1

Edition 1 was deposited as 10.5281/zenodo.23511655. We list what it claimed, what is wrong, and what replaces it. Items 1–4 are blocking: each invalidates a headline claim.

1. **Energy inequality (edition 1, Sec. 2.4 and Theorem 2.5(1)).** Edition 1 argued that $\operatorname{div} u = 0$ gives $\langle (u \cdot \nabla_{\text{eff}})u, u\rangle = 0$, so $E(t) \le E_0$. This is false. With $\nabla_{\text{eff}} = \nabla M$ the nonlinearity is $(u\cdot\nabla)(Mu)$, and skew-symmetry cancels $\langle (u\cdot\nabla)Mu, Mu\rangle$, not $\langle (u\cdot\nabla)Mu, u\rangle$. Section 3 gives exact counterexamples and the Lean proof of the correct statement.

2. **Blow-up claim (Theorem 2.5(1), abstract, discussion).** Edition 1 said the ceiling "caps the vortex-stretching rate at $1/\sqrt{\alpha'}$ and prevents finite-time blow-up for every $\alpha' > 0$". This is withdrawn. $\Omega_{\text{eff}} = \frac{1}{2}\|\nabla Mu\|^2$ is not the enstrophy of $u$, and a bound on a norm with uniformly bounded weights says nothing about the regularity of $u$. No vortex-stretching rate was defined or bounded. For the smooth symbol, the global well-posedness of ML-$\alpha$ is the subject of [6]. For the piecewise symbol, global regularity is open (Section 6).

3. **"Helmholtz Leray-$\alpha$ isomorphism" (Lemma 2.3(1), abstract, Table 1).** The scalar identity $1/(1/|k| + \alpha'|k|) = |k|/(1 + \alpha'|k|^2)$ is true, but it is elementary. The Helmholtz filter's symbol is $1/(1+\alpha'|k|^2)$; $|k|/(1+\alpha'|k|^2)$ is the modulus of the symbol of $\nabla(I-\alpha'\Delta)^{-1}$. Leray-$\alpha$ [3] filters the *advecting* velocity, $(\tilde{u}\cdot\nabla)u$. The model here filters the *advected* field, $(u\cdot\nabla)\tilde{u}$, which is the modified Leray-$\alpha$ model of [6]. That prior work was missing from edition 1. The docstring of `kSmooth` in the shared file `Breakthroughs3DomainsSpec.lean` still carries the old wording. We do not edit that shared file; `ksmooth_eq_mul_weight` in `PaperB_ScaleDualEd2.lean` states the correct relation.

4. **"Certified" computer-assisted proof (Theorem 2.1, Fig. 1, abstract (i)).** All rational arithmetic recomputes correctly (181/500, 93/250, 1473/4000, −33/62500, −2127/4000000, 443/1000, 1/800, 1/200, 1/50). But there is no approximate solution $\bar{u}$, no forcing and no computed residual. $Y_0$, $Z_0$, $Z_1$, $C_{\text{Sob}}$, $C_{\text{alg}}$ and $\|A^\dagger\|$ are posited numbers. The "hurricane boundary-layer" profile has no equations. The Lean theorem checks rational identities and the implication $Y_0 + Z_0 r + Z_2 r^2 < r \Rightarrow Z_0 + Z_2 r < 1$; it is not a fixed-point theorem. `derivedDualScaleTailBound` is a definition, not a theorem. The Lean witness uses $\alpha' = 1/4$ and never constrains it, while the text uses 1/25. The theorem is therefore relabelled *illustrative radii-polynomial arithmetic for hypothetical bounds* (Section 4). The words "certified", "unique fixed point" and "hurricane" are removed.

5. **Tail-bound scaling.** Edition 1 wrote $Z_{\text{tail}} = C_{\text{Sob}}/(\nu M^2)$. The convective term loses one derivative, so the standard estimate scales like $\|\bar{u}\|/(\nu M)$, with the norm of $\bar{u}$ explicit. Only the ratio $1/(\alpha' M^2)$ between the two tails for $M \ge k_*$ is supported, through $\sup_{|k|\ge M} k_{\text{eff}}/|k|^3 = 1/(\alpha' M^4)$ against the Navier–Stokes value $\sup_{|k|\ge M} |k|/|k|^3 = 1/M^2$.

Edition 1 cited Hou and Chen for "$Z_{\text{tail}}$ diverges as $\nu \to 0$". That work concerns Boussinesq/Euler blow-up and makes no such claim; the citation is removed from that sentence, and the authors are Chen and Hou [1, 2].

6. **Kolmogorov exponent (Theorem 2.5(2)).** The "empirical" exponent $-84/125$ had no data, simulation or source; it was a literal in the Lean statement. The Kolmogorov energy-spectrum exponent is $-5/3$, and "$(4/3)(-1/2) = -2/3$" refers to no stated quantity. The item is deleted.

7. $\exp(-1/2)$ **enclosure (Theorem 2.5(3)).** The partial sums were correct, but no Lean theorem mentioned `Real.exp`, and `interval_exp_neg` does not exist. Edition 2 proves $2329/3840 \le e^{-1/2} \le 233/384$ from Mathlib's `Real.exp_bound` (`exp_neg_half_bracket`). The bracket plays no role in any argument.

8. $\mathrm{Sym}^2$ **recurrence and RNS-CRT (Theorem 2.4).** The recurrence is a true algebraic identity, but it has no turbulence content. Shell velocities do not obey a linear second-order recurrence, and a filtered sequence does not obey the same one. The CRT check is vacuous (770 is below each modulus), and $\langle u, v\rangle - \langle v, u\rangle = 0$ is commutativity of the dot product, not convective skew-symmetry. The product was also wrong: $997 \cdot 991 \cdot 983 = 971{,}230{,}541$, not 971,229,691 (Lean: `crt_product_correct`, `crt_product_ed1_wrong`). The Lean theorem also contains Apéry-number recurrences that the paper never mentioned. The identity is moved to Appendix A without physical labels.

9. **"Unique optimal zero-distortion filter" (Lemmas 2.2, 2.3(2)–(3)).** These are elementary properties of the symbol. A function fixed to equal $R$ on $[\sqrt{\alpha'}, \infty)$ and required to be involutive is determined by definition. No dynamical optimality was shown. The multiplier is real, so the smooth filter's defect is amplitude attenuation, not a "dispersive phase error". They are restated as such in Section 2.

10. **Formal-verification claims.** The `leanflow_*` names exist only as `alias` lines for `P2_T1`–`P2_T5`. Edition 1 named a "LeanFlow MCP" architecture; no such server is registered in this repository. The tools actually used are the gate `anse/formal/lean_runner.py` and the harness `sub_projects/socrateai_scientific_agora_leanflow/leanflow_3domains_harness.py`. The edition-1 rejection example ($\kappa = 1.95$) was not shown to run and is removed. `MensuraElenchusThreeMissions.lean` has no `#print axioms` lines, so it fails the gate when run standalone; the lemmas cited from it are covered through the axiom footprints of `P2_T2` and `P2_T3`. The edition-1 abstract said "four" verified contributions while its table listed five.

11. **Parameters.** Edition 1 used $\alpha' = 1/25$ in Theorem 2.1, $1/4$ in Fig. 2 and $1/4$ in the Lean witness. Edition 2 states $\alpha'$ for every number and figure.

12. **Review file in the edition-1 deposit.** The deposit included `breakthrough_3b_navier_stokes_mensura_cap_opus_5round_peer_review.json`, presented as a five-round review with referee findings, scores and an approval verdict. It was produced by a script that calls no language model and no human: the findings were fixed strings, and the scores were pass fractions of local string checks. It was not a review of any kind and is withdrawn. It is not part of edition 2.

## 2 The model and its symbols

On $\mathbb{T}^3 = [0, 2\pi]^3$, write $u(x) = \sum_{k \neq 0} \hat{u}_k e^{ik\cdot x}$ with $k \cdot \hat{u}_k = 0$ and $\hat{u}_{-k} = \overline{\hat{u}_k}$. The scale-dual model is

$$\partial_t u = -P[(u \cdot \nabla)(Mu)] + \nu \Delta u, \qquad \widehat{Mu}(k) = m(|k|^2)\, \hat{u}_k, \tag{1}$$

where $P$ is the Leray projector. Edition 1 wrote the nonlinearity as $(u \cdot \nabla_{\text{eff}})u$ with $\widehat{\nabla_{\text{eff}} u}(k) = i\,(k/|k|)\,k_{\text{eff}}(\alpha', |k|)\,\hat{u}_k$. Since $(k/|k|)k_{\text{eff}} = k\,m$, this is exactly (1) with

$$m_{\text{pw}}(k) = \frac{k_{\text{eff}}(\alpha', |k|)}{|k|} = \min\Big(1, \frac{1}{\alpha'|k|^2}\Big), \qquad m_{\text{ML}}(k) = \frac{k_{\text{smooth}}(\alpha', |k|)}{|k|} = \frac{1}{1+\alpha'|k|^2}. \quad (2)$$

The first identity is `keff_eq_mul_weight` (`PaperB_EnergyBudget.lean`) for the shared definition `kEff`; the second is `ksmooth_eq_mul_weight` (`PaperB_ScaleDualEd2.lean`) for `kSmooth`. With $m_{\text{ML}}$, (1) reads $\partial_t u + (u \cdot \nabla)\tilde{u} + \nabla p = \nu\Delta u$ with $\tilde{u} = (I - \alpha'\Delta)^{-1}u$. This is the ML-$\alpha$ model of [6] (their $\alpha^2$ is our $\alpha'$, and their unfiltered field is our $u$). In Leray-$\alpha$ [3, 5], by contrast, the filtered field is the advecting one.

**Elementary properties of the symbols** (retained from edition 1, Lemmas 2.2–2.3, relabelled). For $a = \alpha' > 0$ and $k > 0$, with $R_{\text{eff}}(a, R) = \max(R, a/R)$ and $R_{\text{smooth}}(a, R) = R + a/R$:

1. $\sqrt{a} \le R_{\text{eff}}(a, R)$, $R_{\text{eff}}(a, a/R) = R_{\text{eff}}(a, R)$, $k_{\text{eff}}(a, 1/(ak)) = k_{\text{eff}}(a, k)$. Also $k_{\text{eff}}(a, k) = k$ for $ak^2 \le 1$ and $k_{\text{eff}}(a, k) = 1/(ak) \le 1/\sqrt{a}$ for $ak^2 \ge 1$.
2. $1/R_{\text{smooth}}(a, 1/k) = k_{\text{smooth}}(a, k)$, $k - k_{\text{smooth}}(a, k) = ak^3/(1 + ak^2) > 0$, and $4ak^2 \le (1 + ak^2)^2$ (so $k_{\text{smooth}} \le 1/(2\sqrt{a})$). The defect is an amplitude attenuation of a real multiplier.
3. If $f(a/R) = f(R)$ for all $R > 0$ and $f(R) = R$ for $R \ge \sqrt{a}$, then $f = R_{\text{eff}}(a, \cdot)$. This follows from the definition and says nothing about dynamical optimality.
4. (New.) $m_{\text{ML}} \le m_{\text{pw}} \le 2\,m_{\text{ML}}$ for $|k| > 0$ (`piecewise_between_smooth`). The two models are therefore spectrally equivalent within a factor 2.

Items 1–3 are the statements `Stmt_P2_T2_MensuraTDualGeometryAndRefutation` and `Stmt_P2_T3_MensuraSmoothAndNoGoTheorem`, proved as `P2_T2` and `P2_T3` in `Breakthroughs3 Domains.lean` (gate: Section 3.3). Figure 1 shows the symbols at $\alpha' = 1/25$.

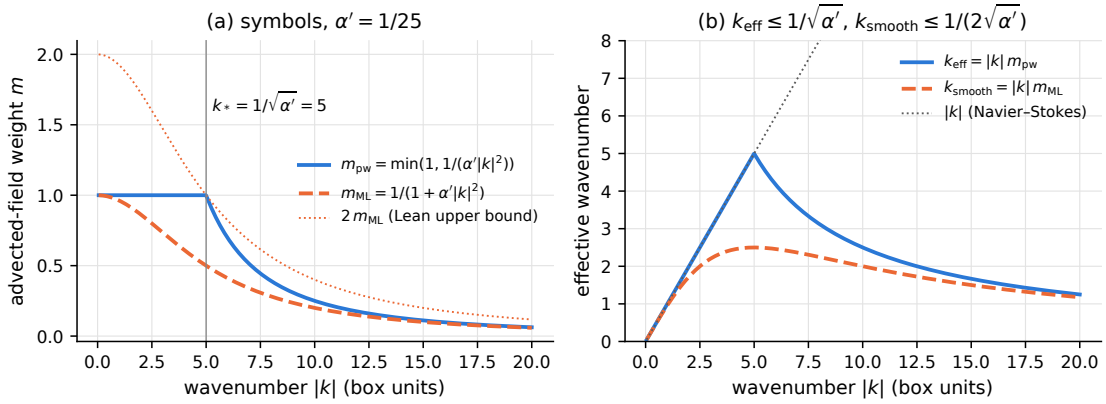



Figure 1: Symbols of the two filters at $\alpha' = 1/25$ ($k_* = 5$), computed from (2). (a) Advected-field weights. The piecewise weight lies between $m_{\text{ML}}$ and $2m_{\text{ML}}$ (Lean `piecewise_between_smooth`). (b) Effective wavenumbers $|k|\,m$ against the Navier–Stokes value $|k|$.

# 3 Contribution: the quadratic invariant of the scale-dual model

## 3.1 Question

Which quadratic quantity does (1) conserve (for $\nu = 0$) or dissipate (for $\nu > 0$)? Does $E(t) \le E(0)$ hold, as edition 1 asserted? How large can $E(t)/E(0)$ become at the paper's own

$(\alpha', \nu) = (1/25, 1/100)$, and which enstrophy-type ceiling survives? This matters because the energy inequality was the only link between the symbol-level algebra of edition 1 and its headline claim of a global enstrophy ceiling.

### 3.2 Lean 4 theorems

Work in any finite Galerkin truncation. In a real mode basis indexed by a finite type $\iota$, the trilinear form of the advection is a tensor $C_{adt}$. Whenever the advecting basis fields are divergence-free it is antisymmetric in its last two slots, $C_{atd} = -C_{adt}$ (`SkewTriad`). The model is $\dot{u}_t = B(u, Mu)_t - \nu\lambda_t u_t$ with $B(u,v)_t = \sum_{a,d} C_{adt} u_a v_d$ and $M = \mathrm{diag}(m)$. The following are proved in `formal/ANSE/PaperB_EnergyBudget.lean` (namespace `ANSE.PaperB_EnergyBudget`) and `formal/ANSE/PaperB_ScaleDualEd2.lean` (namespace `ANSE.PaperB_ScaleDualEd2`). Each is quantified over the index type, the tensor, the weights and the state.

**Theorem 3.1** (Invariant and no-go)**.** *Let $C$ be skew, $m, u : \iota \to \mathbb{R}$.*

1. *(`weighted_transfer_zero`) $\sum_t m_t u_t\, B(u, Mu)_t = 0$. Thus $Q = \frac{1}{2}\sum_t m_t u_t^2$ receives no nonlinear transfer.*
2. *(`l2_transfer_half_difference`) $\sum_t u_t\, B(u, Mu)_t = \frac{1}{2}\sum_a u_a \sum_{d,t}(m_d - m_t) C_{adt} u_d u_t$. The $L^2$ energy moves only through weight differences inside a triad.*
3. *(`l2_conservation_iff_constant`) $(\forall C \text{ skew},\ \forall u :\ \sum_t u_t B(u, Mu)_t = 0) \iff \forall d, t :\ m_d = m_t$.*
4. *(`weighted_energy_hasDerivAt`, `weighted_energy_antitone`) Along any trajectory with $\dot{u}_i = B(u, Mu)_i - \nu\lambda_i u_i$, $\frac{d}{dt}Q = -\nu\sum_i m_i\lambda_i u_i^2$. Hence $Q$ is antitone whenever $m, \lambda \geq 0$ and $\nu \geq 0$.*

**Theorem 3.2** (Corrected ceiling)**.** *Let $a = \alpha' > 0$, $k_i^2 > 0$ and $A_i = |\hat{u}_i|^2 \geq 0$.*

1. *(`omega_eff_le_Q_div_alpha`) $\frac{1}{2}\sum_i k_i^2 m_{\mathrm{pw},i}^2 A_i \leq \frac{1}{a}\cdot\frac{1}{2}\sum_i m_{\mathrm{pw},i} A_i$, i.e. $\Omega_{\mathrm{eff}} \leq Q/\alpha'$ for the piecewise symbol.*
2. *(`omega_smooth_le_Q_div_alpha`) The same inequality for $m_{\mathrm{ML}}$, for $k_i^2 \geq 0$.*
3. *(`Q_le_E`) $\sum_i m_{\mathrm{pw},i} A_i \leq \sum_i A_i$, i.e. $Q \leq E$. (For $m_{\mathrm{ML}} \leq m_{\mathrm{pw}}$, see item 4 of Section 2.)*

*Together with Theorem 3.1(4): $\Omega_{\mathrm{eff}}(t) \leq Q(t)/\alpha' \leq Q_0/\alpha' \leq E_0/\alpha'$ along every Galerkin trajectory.*

**Proposition 3.3** (Exact growth threshold; `growth_threshold_iff`, `growth_at_34`, `decay_at_33`)**.** *For $\lambda > 0$: $\lambda^3 \cdot \frac{6}{13} - \lambda^2 \cdot \frac{78}{5} > 0 \iff \lambda > \frac{169}{5}$. At $\lambda = 34$ the value is $6936/65$; at $\lambda = 33$ it is negative.*

The numbers $6/13$ and $78/5 = 2\nu\Omega$ are the exact nonlinear rate and the viscous rate of the $\alpha' = 1/25$ triad of Section 3.4 at $\nu = 1/100$. The identification of the polynomial with $dE/dt$ comes from the exact computation, not from Lean.

**Controls in Lean.** Positive: `ns_energy_conserved` ($m \equiv 1$ conserves $E$) and `ns_weight_one_is_smooth_at_zero`. Negative: `skew_needed` (without skew-symmetry the weighted transfer is non-zero), `ns_enstrophy_not_capped` (with $m \equiv 1$ no ceiling $k^2 \leq 1/\alpha'$ holds) and `decay_at_33`. Also proved: `exp_neg_half_bracket`, $2329/3840 \leq$ `Real.exp`$(-1/2) \leq 233/384$, from `Real.exp_bound` with seven terms.

### 3.3 Gate result

The only accepted gate is `.venv/bin/python -m anse.formal.lean_runner formal/ANSE/<File>.lean`, run from the repository root. It compiles the file, requires a `#print axioms` line for every theorem, rejects `sorryAx`, and rejects any axiom outside {`propext`, `Classical.choice`, `Quot.sound`}. The output was:

| File | exit | result |
|---|---|---|
| `PaperB_EnergyBudget.lean` (new) | 0 | 13/13 compiled; axioms {propext, Classical.choice, Quot.sound}; no sorryAx |
| `PaperB_ScaleDualEd2.lean` (new) | 0 | 12/12 compiled; axioms ⊆ {propext, Classical.choice, Quot.sound} (the two CRT lemmas use only propext); no sorryAx |
| `Breakthroughs3Domains.lean` (shared) | 0 | 26/26 compiled and clean; contains `P2_T1`–`P2_T5` |
| `Breakthroughs3DomainsSpec.lean` (shared) | 1 | 1 entry, `compiled: false`, `has_sorry: true` — the runner's "unchecked" sentinel; declares no theorem (definitions and `Prop` statements only) |
| `MensuraElenchusThreeMissions.lean` | 1 | 18 entries, `compiled: false`, `has_sorry: true` — all the same "no `#print axioms` line: unchecked" sentinel |

The last two rows are transcribed literally from the gate, and the sentinel needs reading carefully: `anse/formal/lean_runner.py` reports a declaration with no `#print axioms` line as `compiled: false, has_sorry: true`, which is how it refuses to certify what it could not check. The `Breakthroughs3DomainsSpec.lean` row carries the same two flags for a different reason: that file declares no theorem at all, so the runner returns a single such entry for the file rather than one per declaration. It is *not* a finding of `sorry`: neither file contains a `sorry` term (a literal `grep` does match once, inside a `Breakthroughs3DomainsSpec.lean` docstring that says "0 `sorryAx`"), and both compile transitively, since `Breakthroughs3Domains.lean` imports them and passes 26/26. We therefore claim nothing for the declarations in these two files; `P2_T2` and `P2_T3`, which we do cite, live in the file that passes. What the shared theorems assert, exactly: `P2_T1` proves rational identities and the implication above (no PDE). `P2_T2` and `P2_T3` prove items 1–3 of Section 2. `P2_T4` proves the $\mathrm{Sym}^2$ identity, the Apéry recurrence values $5, 73, 1445$ and the integer dot product 770. `P2_T5` proves a three-variable inequality $k_i^2 \le 1/a \Rightarrow \sum k_i^2 u_i^2 \le \frac{1}{a}\sum u_i^2$ and rational partial sums; it contains no time, PDE or `Real.exp`.

### 3.4 Exact computation

All quantities below are computed over $\mathbb{Q}(i)$ with Python integers and `Fractions`, with no floating point. We use $dE/dt = -\langle u, (u\cdot\nabla)Mu\rangle$ and $dQ/dt = -\langle Mu, (u\cdot\nabla)Mu\rangle$ (volume means), and the projector drops out because $u$ and $Mu$ are solenoidal.

**A1: triad at $\alpha' = 1/4$ ($k_* = 2$).** Take modes $p = (1,0,0)$, $q = (0,2,0)$, $r = (-1,-2,0)$ with amplitudes $(0,1,1)$, $(1,0,i)$, $(2,-1,1+i)$ and their conjugates. Then $E_0 = 11$ and $\Omega = 45$, and:

| model | $Q_0$ | $dE/dt$ (inviscid) | $dQ/dt$ (inviscid) | $dE/dt$ at $\nu = 1/100$ |
|---|---|---|---|---|
| Navier–Stokes ($m \equiv 1$) | 11 | 0 | 0 | $-9/10$ |
| piecewise $m_{\mathrm{pw}}$ | 48/5 | $+6/5$ | 0 | $+3/10$ |
| smooth $m_{\mathrm{ML}}$ | 257/45 | $+32/15$ | 0 | $+37/30$ |

At edition 1's own viscosity the model's kinetic energy therefore *increases* at this instant. Its invariant $Q$ does not.

**A2: triad at the paper's $\alpha' = 1/25$ ($k_* = 5$).** Take modes $(5,0,0)$, $(0,1,0)$, $(-5,-1,0)$, with $|k|^2 = 25, 1, 26$ and weights $m_{\mathrm{pw}} = 1, 1, 25/26$, and amplitudes $(0,1,1)$, $(1,0,i)$, $(1,-5,1+i)$. Then $E_0 = 32$, $Q_0 = 402/13$, $\Omega = 780$, nonlinear $dE/dt = R = 6/13$ and $dQ/dt = 0$. Scaling the field by $\lambda$ gives $dE/dt = \lambda^3 R - 2\nu\lambda^2\Omega$. At $\nu = 1/100$ this is positive exactly for $\lambda > \lambda^* = 2\nu\Omega/R = 169/5$; at $\lambda = 34$ it equals $+6936/65$ (Proposition 3.3). The threshold is $\lambda^*(\nu) = 3380\,\nu$ (Fig. 4(c)). The Navier–Stokes control on the same field gives $dE/dt = 0$.

**A3: sweep over triads.** Enumerate every triad $p + q + r = 0$ with $p$ in the fundamental domain $p_1 \geq p_2 \geq p_3 \geq 0$ of the cube group, all $|k|^2 \leq K$, and six distinct modes. Give each mode an integer, exactly solenoidal amplitude $\hat{u}_k = k \times a + i\, k \times b$, with $a, b \in \{-3, \ldots, 3\}^3$ drawn from a seeded generator. The counts below are of *ordered* enumerations, not of geometrically distinct triads: the loop runs $p$ over the fundamental domain and $q$ over the whole lattice, so $(p,q,r)$ and $(p,r,q)$ are counted separately with independently drawn amplitudes, and a triad with two members in the fundamental domain is reached twice. Each row is therefore best read as a number of amplitude draws. The $dQ/dt = 0$ and $dE/dt \neq 0$ conclusions do not depend on the convention. The results are:

| $\alpha'$, $K$ | model | ordered triads $(p,q,r)$ | non-constant $m$ | $dE/dt \neq 0$ | $dQ/dt \neq 0$ |
|---|---|---|---|---|---|
| 1/4, 12 | NS | 1,039 | 0 | 0 | 0 |
| | piecewise | 1,039 | 983 | 972 | 0 |
| | smooth | 1,039 | 1,029 | 1,017 | 0 |
| 1/25, 36 | NS | 18,886 | 0 | 0 | 0 |
| | piecewise | 18,886 | 12,241 | 12,200 | 0 |
| | smooth | 18,886 | 18,854 | 18,775 | 0 |

In every model triad $dQ/dt = 0$ exactly. No triad with constant weights moves $L^2$ energy, and almost every triad with non-constant weights does. This is the content of Theorem 3.1(2)–(3), observed in exact arithmetic. The few non-constant-weight triads with $dE/dt = 0$ are cancellations for the drawn amplitudes.

**Where the energy is created.** Figure 2 shows, for a genuinely three-dimensional triad at $\alpha' = 1/4$, the local density $-u \cdot [(u \cdot \nabla)(Mu)]$ whose volume mean is $dE/dt$. The modes are $(1,1,0)$, $(0,1,1)$, $(-1,-2,-1)$ with amplitudes $k \times a + i\, k \times b$; the field is listed in `paper_b_v2_fig2_values.json`. For this field $E_0 = 14$ and $Q_0 = 34/3$, with exact rates $dE/dt = 8/3$ and $dQ/dt = 0$. The $48^3$ grid mean is 2.6666666666666625. For Navier–Stokes the density is $-u \cdot \nabla(|u|^2/2)$, a divergence, and its mean is $5 \times 10^{-16}$ (exact 0). The density itself is not unique, since divergence terms can be added; only its mean is physical.

### 3.5 Pseudospectral Galerkin ensemble

**Method.** The solver is a 2/3-dealiased Fourier–Galerkin discretization of (1), advanced with RK4 and an exact viscous integrating factor, on $N^3$ grids (`scripts/three_breakthroughs/paper_b_v2_experiment.py`). Initial data are random-phase solenoidal fields with spectrum $\propto k^4 e^{-2(k/4)^2}$, normalized to energy $E_0$. The pre-registered ensemble uses $N = 32$, $dt = 0.005$, $t \leq 3$, ten seeds, the three models, $\alpha' \in \{1/4, 1/16, 1/25\}$ and $\nu \in \{0, 1/400, 1/100\}$, at $E_0 = 1/2$. Convergence checks at the paper's $(\alpha', \nu)$ use $N = 64$ (zero-padded, identical initial data) and $dt/2$. A 0.5-time-unit smoke run of one seed at $E_0 = 1/2$ showed $E$ decreasing. Before any ensemble result existed, we therefore added an amplitude scan $E_0 \in \{2, 8, 32\}$ (5 seeds) at the paper's $(\alpha', \nu)$, with convergence checks at $E_0 = 32$. The amendment is recorded in the script's docstring. In total there are 299 runs (13,257 s wall time on 7 processes). One caveat on "pre-registered": the criteria, the controls and the amendment were all fixed in that docstring

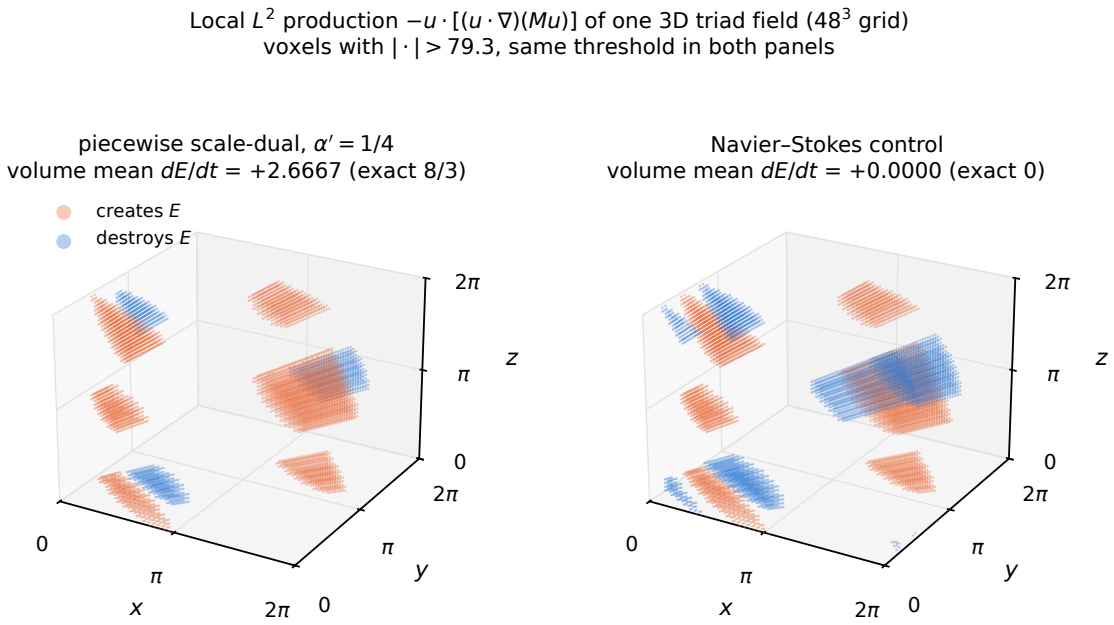



Figure 2: Local $L^2$ production density on $\mathbb{T}^3$ for one exact solenoidal triad at $\alpha' = 1/4$, recomputed from the model on a $48^3$ grid. Orange voxels create kinetic energy and blue voxels destroy it; both panels use the same threshold, the 93rd percentile of $|\cdot|$ for the scale-dual field. Left: piecewise scale-dual model, volume mean $+8/3$. Right: Navier–Stokes control on the same field, volume mean 0.

before the runs they govern, but the script was untracked in git at the time, so the ordering rests on the author's record and on file timestamps (the script was last written at 08:24:53 and the results JSON at 12:12:06, after a 13,257 s run that therefore began around 08:31), not on a commit predating the run.

**Code validation.** On the exact fields of A1 and A2, the float rates agree with the exact rationals 0, 6/5, 32/15 and 6/13 to $1.1 \times 10^{-15}$ at $N = 16$ and 32.

**Ensemble ($N = 32$, $E_0 = 1/2$, ten seeds; mean $\pm$ s.d. of $\max_{t\le 3} E/E_0$).**

| model | $\alpha'$ | $\nu = 0$ | $\nu = 1/400$ | $\nu = 1/100$ |
|---|---|---|---|---|
| Navier–Stokes | – | 1.0000 (drift $7.5 \times 10^{-10}$) | 1 | 1 |
| piecewise | 1/4 | $1.459 \pm 0.022$ | 1 | 1 |
| | 1/16 | $2.892 \pm 0.034$ | $1.396 \pm 0.018$ | 1 |
| | 1/25 | $2.722 \pm 0.021$ | $1.361 \pm 0.011$ | 1 |
| smooth (ML-$\alpha$) | 1/4 | $1.249 \pm 0.012$ | 1 | 1 |
| | 1/16 | $1.814 \pm 0.018$ | 1 | 1 |
| | 1/25 | $1.865 \pm 0.017$ | $1.007 \pm 0.006$ (8/10 seeds $> 1$) | 1 |

An entry 1 means $E$ never exceeded $E_0$ (its maximum is the initial value). In every inviscid model run $Q$ was conserved to $|Q/Q_0 - 1| \le 9.5 \times 10^{-11}$; the bound over the ensemble group alone is $8.7 \times 10^{-12}$, and the looser figure comes from the inviscid $N = 64$ piecewise run (the triad evolution gives $4.3 \times 10^{-12}$). In every viscous model run $Q$ never increased (largest increase 0). In every model run $\max_t \alpha'\Omega_{\text{eff}}/Q_0 \le 0.9967 \le 1$, as Theorem 3.2 requires. The Galerkin bound $E \le \max(1, \alpha' N_{\max}^2)\, Q$ held in all runs. Figure 3 shows the trajectories in $(E/E_0, Q/Q_0, \alpha'\Omega_{\text{eff}}/Q_0)$ space, and Fig. 4(a,b) the ensemble time series at $\alpha' = 1/25$.

**Exact triad field, evolved ($\alpha' = 1/4$, $N = 32$, $dt = 0.002$, $t \leq 2$).** $\max_t E/E_0 = 2.511$ for piecewise and 1.539 for smooth when inviscid ($Q$ conserved to $4.3 \times 10^{-12}$). At $\nu = 1/100$ they are 1.510 and 1.138, both at $t \approx 1.22$. Navier–Stokes stays at 1.

**Resolution dependence (inviscid, piecewise, $\alpha' = 1/25$, seed 0).** $\max_t E/E_0 = 2.707$ at $N = 32$ and 8.396 at $N = 64$. Energy growth is a transfer of $Q$-conserving content to subgrid modes, whose number grows with $N$ (Section 6, item 4).

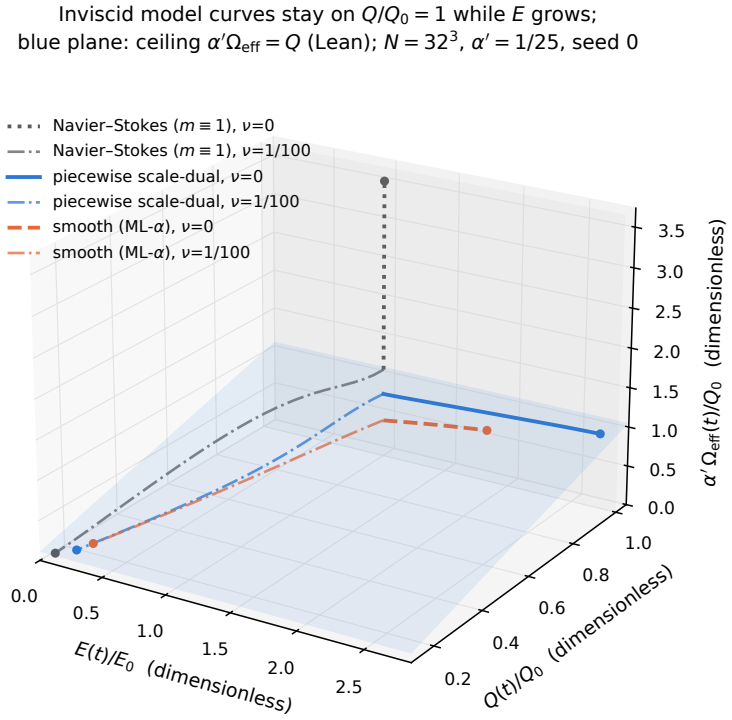



Figure 3: Trajectories of seed 0 ($N = 32^3$, $\alpha' = 1/25$, $E_0 = 1/2$, $t \leq 3$) in the space of kinetic energy $E/E_0$, invariant $Q/Q_0$ and rescaled effective enstrophy $\alpha'\Omega_{\text{eff}}/Q_0$. All quantities are dimensionless ratios. Inviscid model curves (solid, dashed) stay on the plane $Q/Q_0 = 1$ (grey) while $E/E_0$ rises to $\approx 2.7$ (piecewise) and $\approx 1.8$ (smooth). The inviscid Navier–Stokes curve (dotted) stays on $E/E_0 = 1$, and its $\alpha'\Omega/E_0$ rises to 3.6, above the ceiling. The translucent blue plane is the Lean ceiling $\alpha'\Omega_{\text{eff}} = Q$. Model curves never cross it. Dash-dot curves: $\nu = 1/100$. Dots mark $t = 3$.

### 3.6 Controls

We fixed the controls before running the model numbers. Both sets passed (`summary.positive_controls.passed` and `summary.negative_controls.passed` in the results file).

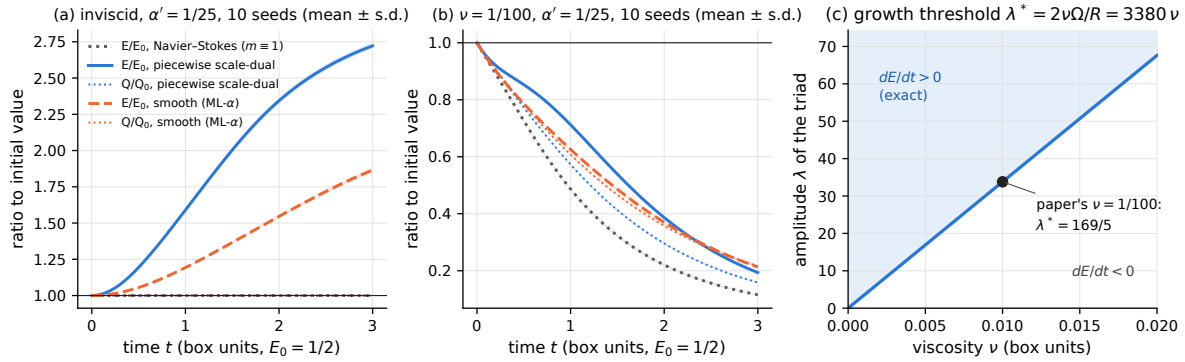



Figure 4: (a,b) $E/E_0$ (thick) and $Q/Q_0$ (thin dotted) against time, mean ± one standard deviation over 10 seeds ($N = 32^3$, $\alpha' = 1/25$, $E_0 = 1/2$). In (a) the inviscid $Q/Q_0$ curves coincide with the line 1. (c) Exact growth threshold of the $\alpha' = 1/25$ triad of Section 3.4: $dE/dt > 0$ exactly when the amplitude exceeds $\lambda^*(\nu) = 3380\,\nu$; edition 1's $\nu = 1/100$ gives $\lambda^* = 169/5$ (Lean `growth_threshold_iff`).

| type | check | observed |
|---|---|---|
| positive | exact NS triads (A1, A2): $dE/dt = dQ/dt = 0$ | 0, 0 |
| positive | exact sweeps: NS triads with $dE/dt \neq 0$; constant-weight triads with $dE/dt \neq 0$ | 0; 0 |
| positive | float rates equal the exact rationals | max error $1.1 \times 10^{-15}$ |
| positive | inviscid dealiased NS: max $\lvert E/E_0 - 1\rvert$ over 10 seeds (RK4 time error) | $7.5 \times 10^{-10}$ |
| positive | Lean ceiling: max $\alpha'\Omega_{\text{eff}}/Q_0$ over all model runs; $\max(\alpha'\Omega_{\text{eff}} - Q)/Q_0$ | 0.9967; −0.0013 |
| negative | exact NS field with one non-solenoidal mode ($k \cdot \hat{u}_k = 1$) | $dE/dt = -2 \neq 0$ |
| negative | aliased (not dealiased) inviscid NS, seed 0: max $\lvert E/E_0 - 1\rvert$ | 4.6 at the last finite sample $t = 0.90$, then overflow between $t = 0.90$ and $t = 0.95$; dealiased: $7.3 \times 10^{-10}$ |
| negative | NS runs whose true enstrophy exceeds $E_0/\alpha'$ (no ceiling without the filter; Lean `ns_enstrophy_not_capped`) | 30 of 30 |

The last line is also the empirical form of item 2 in Section 1. For the scale-dual models the true enstrophy is not capped either: $\max_t \alpha'\Omega/E_0$ reached 1.1–17 across the ensemble, while $\alpha'\Omega_{\text{eff}}/Q_0$ stayed below 1.

### 3.7 Outcome

**Edition-2 claim.** The claim is: "$E$ is not a Lyapunov functional of the scale-dual model; $Q$ is." The pre-registered falsifiers were any exact divergence-free field with $dQ/dt \neq 0$, or no non-constant-weight triad with $dE/dt \neq 0$. Neither occurred: there were 0 fields with $dQ/dt \neq 0$ out of 59,780 exact evaluations (A1, A2, and both sweeps for all three models), and 12,200 ($\alpha' = 1/25$) and 972 ($\alpha' = 1/4$) piecewise triads move $L^2$ energy. **Not falsified.**

**Edition-1 Theorem 2.5, $E(t) \leq E_0$, at $(\alpha', \nu) = (1/25, 1/100)$.** The pre-registered refutation needs a positive exact rate for some amplitude (A2: yes, for $\lambda > 169/5$). It also needs, in the DNS, $\max_t E/E_0 - 1$ above 10× the NS control's excess (floored at $10^{-12}$), persisting under $N \to 64$ and $dt \to dt/2$. Results (Fig. 5):

- $E_0 = 1/2$: $E(t) \leq E_0$ in all 10 seeds and in the $N = 64$ and $dt/2$ reruns, for both models. At this amplitude only instantaneous growth (the exact triads) is shown.

- $E_0 = 2$ and 8: the piecewise model exceeds $E_0$ in 5/5 seeds ($\max E/E_0 = 1.034 \pm 0.006$ and $1.356 \pm 0.008$). The smooth model exceeds it at $E_0 = 8$ in 4/5 seeds (up to 1.012).
- $E_0 = 32$: piecewise $1.767 \pm 0.015$, smooth $1.247 \pm 0.014$ (5/5 seeds each). For seeds 0–2, halving $dt$ changes $\max E/E_0$ by $3.5 \times 10^{-7}$ (piecewise) and $\leq 3.8 \times 10^{-8}$ (smooth). $N = 64$ *increases* it to 2.96–3.00 (piecewise) and 1.50–1.52 (smooth). The NS control stays at exactly 1.

**Verdict: refuted for the piecewise model at** $E_0 = 32$ by the pre-registered criterion, and also exceeded for the smooth (ML-$\alpha$) model. The excess is robust to the time step but its magnitude is not converged in $N$. It increases with resolution, consistent with energy transfer to subgrid modes permitted by $Q$-conservation. We therefore claim that $E(t) > E_0$ occurs and persists under refinement at the paper's parameters; we do not claim a converged value of $\sup_t E/E_0$.

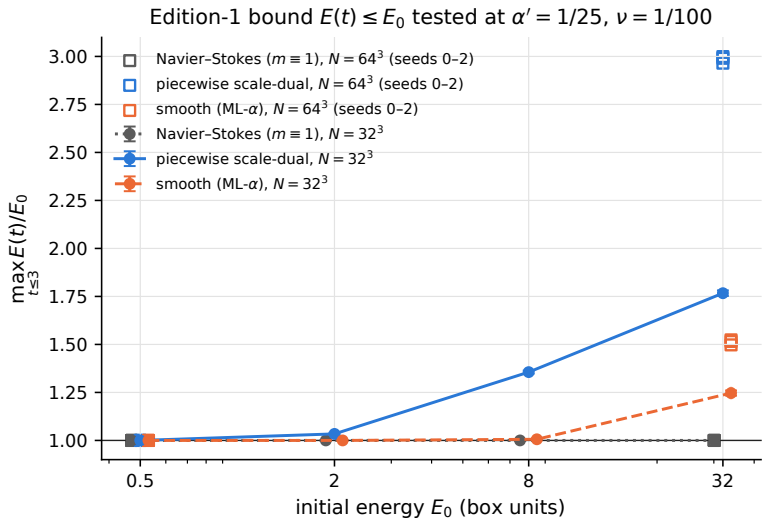



Figure 5: Test of edition 1's $E(t) \leq E_0$ at its own $\alpha' = 1/25$, $\nu = 1/100$. The plot shows $\max_{t \leq 3} E/E_0$ against the initial energy $E_0$. Filled markers: $N = 32^3$ mean $\pm$ s.d. (10 seeds at $E_0 = 1/2$, 5 seeds otherwise), with individual seeds as faint dots. Open squares: $N = 64^3$ reruns of seeds 0–2 at $E_0 = 1/2$ and 32. Navier–Stokes stays at 1.

## 4 Retained arithmetic from edition 1, relabelled

**Illustrative radii-polynomial arithmetic (edition 1, Theorem 2.1).** Take the *hypothetical* bounds $Y_0 = 1/10000$, $Z_0 = 7/20$, $Z_1 = 1/200$, $Z_{\text{tail}} = 1/200$, $Z_2 = 12$ and $r_* = 1/1000$. Then $Z_0 + Z_2 r_* = 181/500$, $Z_0 + Z_1 + Z_{\text{tail}} + Z_2 r_* = 93/250$, and $P(r_*) = Y_0 + (Z_0 + Z_1 + Z_{\text{tail}} - 1)r_* + Z_2 r_*^2 = -33/62500 < 0$. Replacing $Z_{\text{tail}}$ by $1/800$ gives $1473/4000$ and $-2127/4000000$. These identities are proved over $\mathbb{Q}$ in `P2_T1`. If such bounds were *derived* for an actual approximate solution, the radii-polynomial theorem [4, 7] would give a zero of $F$ within $r_*$. No such derivation exists here; Section 6, item 3 describes what one would require. The former "hurricane" number $443/1000$ is the sum $21/50 + 3/500 + 1/500 + 15/1000$ and refers to no equation.

## 5 Scope and limitations

- The Lean theorems concern finite Galerkin truncations with an abstract skew tensor. They do not formalize $\mathbb{T}^3$, the Fourier basis or the Leray projector. That the Fourier–Galerkin

truncation of (1) has this form is standard but not machine-checked. Passing to the limit $N \to \infty$ (existence and regularity of solutions) is not addressed.

- The ceiling $\Omega_{\text{eff}} \le E_0/\alpha'$ bounds a model-internal quantity. $\Omega_{\text{eff}} = \frac{1}{2}\|\nabla M u\|^2$ is not the enstrophy of $u$, and no statement about blow-up or regularity of (1) or of Navier–Stokes follows from it. In our runs the true enstrophy is not bounded by $E_0/\alpha'$ (Section 3.6).

- The DNS are $32^3$ (and $64^3$ checks) on a $2\pi$-periodic box over $t \le 3$, with one random-phase spectral shape. They are not low-Reynolds-number runs, and we do not claim they are: at the refutation amplitude $E_0 = 32$ with $\nu = 1/100$, $u_{\text{rms}} \approx 4.6$ and $L \approx 1.6$ give a Reynolds number of order $7 \times 10^2$ and a Kolmogorov wavenumber of order $10^2$, far above $k_{\max} = 10$ ($N = 32$) or 21 ($N = 64$). The $E_0 = 32$ runs are strongly under-resolved, which is consistent with doubling $N$ raising $\max E/E_0$ from about 1.77 to about 3.0 (two resolutions cannot establish that the under-resolution is the cause); the occurrence and persistence of $E > E_0$ is what we claim from them, never a converged value. "Units" throughout are box units ($L = 2\pi$, unit density). They test the energy budget. They are not turbulence simulations, and inviscid runs of a truncated system thermalize.

- The sign of $dE/dt$ is cubic in $u$ and flips under $u \to -u$. The "$dE/dt > 0$" counts in the sweep are therefore not growth probabilities; the relevant count is the number of triads that move $L^2$ energy at all.

- No PARI/GP or Sage installation was available on the build host. All exact arithmetic here is integer/rational arithmetic in Python (`fractions`) and in the Lean kernel (`norm_num`); it involves no number theory.

- This edition has not been reviewed by anyone outside the author's own checks.

## 6 Research directions

1. *Global regularity for the piecewise symbol.* By $m_{\text{ML}} \le m_{\text{pw}} \le 2m_{\text{ML}}$, the invariant $Q$ is equivalent, within a factor 2, to the ML-$\alpha$ energy $\frac{1}{2}\langle u, \tilde{u}\rangle = \frac{1}{2}(\|\tilde{u}\|^2 + \alpha'\|\nabla\tilde{u}\|^2)$, $\tilde{u} = (I - \alpha'\Delta)^{-1}u$. Does the ML-$\alpha$ well-posedness argument [6] carry over despite the kink of $m_{\text{pw}}$ at $k_*$? This is open until proved.

2. *Dynamical test of "zero inertial distortion".* At $64^3$ and above, compare resolved-band trajectories and spectra of the piecewise model and ML-$\alpha$ against a Navier–Stokes reference at the same $\nu$. The kink may instead cause Gibbs-type artefacts.

3. *A genuine computer-assisted proof.* Validate a steady state of Kolmogorov-forced ML-$\alpha$ or scale-dual flow on $\mathbb{T}^3$. This needs a computed $\bar{u}$, interval-arithmetic $Y_0, Z_0, Z_1, Z_2$, and a tail bound that keeps the one-derivative loss and $\|\bar{u}\|$ explicit. Only the final rational inequalities would go to Lean. Chen and Hou [1, 2] show what such a fluid CAP involves.

4. *Energy transfer to subgrid scales.* In a truncation $|k| \le N$ with $N > k_*$, $E \le \alpha' N^2 Q$ (checked in every run). Measure how $\sup_t E/E_0$ scales with $N$: Section 3.5 gives the first two points.

5. *Spectral scaling beyond $k_*$.* Measure the piecewise model's spectrum and compare it with the published Leray-$\alpha$ and ML-$\alpha$ predictions. This would replace the deleted "$-84/125$".

## Reproducibility

From the repository root:

- `.venv/bin/python scripts/three_breakthroughs/paper_b_v2_experiment.py` (writes `papers/three_breakthroughs/paper_b/paper_b_v2_results.json`);
- `.venv/bin/python scripts/three_breakthroughs/paper_b_v2_figures.py` (writes the figure PDFs and `paper_b_v2_fig2_values.json`);
- `.venv/bin/python -m anse.formal.lean_runner formal/ANSE/PaperB_EnergyBudget.lean`, and the same for `formal/ANSE/PaperB_ScaleDualEd2.lean`;
- `.venv/bin/python -m pytest tests/test_paper_b_v2_experiment.py -q -p no:cacheprovider`.

The experiment's design, controls and falsification criteria are fixed in the docstring of the experiment script. That includes the one amendment made before any ensemble result existed.

One naming warning for anyone reading the bundle: the figure *file* names do not follow the figure numbers in this PDF. Figure 1 is `fig5_v2_symbols.pdf`, Figure 2 is `fig2_v2_energy_production_density_3d.pdf`, Figure 3 is `fig1_v2_invariant_trajectories_3d.pdf`, Figure 4 is `fig3_v2_energy_vs_Q_and_threshold.pdf` and Figure 5 is `fig4_v2_amplitude_scan.pdf`. The numbering inside the document is consistent; only the file names are historical.

## A An algebraic identity (edition 1, Theorem 2.4)

If $u_{n+2} = au_{n+1} + bu_n$ and $v_{n+2} = av_{n+1} + bv_n$, then $w_n = u_n v_n$ satisfies $w_{n+3} = (a^2+b)w_{n+2} + b(a^2+b)w_{n+1} - b^3 w_n$. This is the first conjunct of `P2_T4`, proved by `ring`. No physical interpretation is claimed.

## References


[1] J. Chen and T. Y. Hou, "Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: Analysis," `arXiv:2210.07191`.

[2] J. Chen and T. Y. Hou, "Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data II: Rigorous numerics," `arXiv:2305.05660`.

[3] A. Cheskidov, D. D. Holm, E. Olson, and E. S. Titi, "On a Leray-$\alpha$ model of turbulence," *Proc. R. Soc. A*, vol. 461, pp. 629–649, 2005.

[4] S. Day, J.-P. Lessard, and K. Mischaikow, "Validated continuation for equilibria of PDEs," *SIAM J. Numer. Anal.*, vol. 45, no. 4, pp. 1398–1424, 2007.

[5] C. Foias, D. D. Holm, and E. S. Titi, "The Navier–Stokes-alpha model of fluid turbulence," *Physica D*, vol. 152–153, pp. 505–519, 2001.

[6] A. A. Ilyin, E. M. Lunasin, and E. S. Titi, "A modified-Leray-$\alpha$ subgrid scale model of turbulence," *Nonlinearity*, vol. 19, no. 4, pp. 879–897, 2006. DOI: 10.1088/0951-7715/19/4/006.

[7] M. T. Nakao, M. Plum, and Y. Watanabe, *Numerical Verification Methods and Computer-Assisted Proofs for Partial Differential Equations*. Springer, 2019.